

1 Lecture 20-21

1.1 Topics

- Infinite Series
- Telescoping Series, Geometric Series
- Algebraic Limit Theorem for Series
- Divergence Test
- Cauchy Criterion for Series
- Absolute Convergence Test

1.2 Infinite Series

Consider the following expression:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

How can we make sense of the infinite sum above? More generally, let (a_n) be a sequence of real numbers. Then what does the following expression mean?

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots ?$$

Definition (Infinite Series). Let $(X, \|\cdot\|)$ be a normed space. Let (x_n) be a sequence in X .

(*) An expression of the form

$$\sum_{n=1}^{\infty} x_n = x_1 + x_2 + x_3 + \cdots$$

is called an **infinite series**.

(*) $x_1, x_2, \dots$ are called the **terms** of this infinite series.

(*) The corresponding sequence of **partial sums** is defined by

$$\forall m \in \mathbb{N} \quad s_m = (\text{finite}) \text{ sum of the first } m \text{ terms of the series;}$$

that is,

$$\begin{aligned} s_1 &= x_1 \\ s_2 &= x_1 + x_2 \\ s_3 &= x_1 + x_2 + x_3 \\ &\vdots \\ s_m &= x_1 + x_2 + \cdots + x_m \\ &\vdots \end{aligned}$$

1. We say that the infinite series $\sum_{n=1}^{\infty} x_n$ converges to $L \in X$ (and we write $\sum_{n=1}^{\infty} x_n = L$) if $\lim_{n \rightarrow \infty} s_m = L$.

2. We say that the infinite series **diverges**, if (s_m) diverges.

3. If $X = \mathbb{R}$ and $s_m \rightarrow \infty$, we write $\sum_{n=1}^{\infty} x_n = \infty$.

(*) If $X = \mathbb{R}$ and $s_m \rightarrow -\infty$, we write $\sum_{n=1}^{\infty} x_n = -\infty$.

Remark (1). Given an infinite series $\sum_{n=1}^{\infty} x_n$, it is important to keep a clear distinction between

- (a) the sequence of terms: $(x_1, x_2, x_3, \dots)$
- (b) the sequence of partial sums: $(s_1, s_2, s_3, \dots)$.

Remark (2). We may sometimes consider infinite series where the summation begins with $n = 0$ or $n = n_0$ for some integer n_0 different from 1.

As we shall see, some of our theorems apply specifically to series in $\mathbb{R}$ or to series with terms in $[0, \infty)$. Also, in our examples, we will primarily focus on series in $\mathbb{R}$; however, we will also consider encounter highly useful theorems that hold in more general normed spaces.

In most cases it is difficult (or even impossible) to find a simple formula for the partial sum s_m . However, there are two types of series for which we can easily find a simple formula for the partial sums. These two types are:

- (1) Telescoping Series
- (2) Geometric Series

1.3 Telescoping Series, Geometric Series

Example. Consider the following series

$$\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right).$$

Notice that $x_n = \frac{1}{n} - \frac{1}{n+1}$. The corresponding sequence of partial sums is

$$\begin{aligned} s_1 &= 1 - \frac{1}{2} \\ s_2 &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) = 1 - \frac{1}{3} \\ s_3 &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) = 1 - \frac{1}{4} \\ &\vdots \\ s_m &= \sum_{n=1}^m \left(\frac{1}{n} - \frac{1}{n+1} \right) = \left(\sum_{n=1}^m \frac{1}{n} \right) - \left(\sum_{n=1}^m \frac{1}{n+1} \right) \\ &= 1 - \frac{1}{m+1}. \end{aligned}$$

Clearly, we see that

$$\lim_{m \rightarrow \infty} s_m = \lim_{m \rightarrow \infty} \left[1 - \frac{1}{m+1} \right] = 1.$$

Hence, $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges to 1.

In general, a telescoping series is an infinite series where partial sums eventually have a finite number of terms after cancellation. For example, if (y_n) is a sequence in the normed space $(X, \|\cdot\|)$, then $\sum_{n=1}^{\infty} (y_n - y_{n+1})$ is a telescoping series; that is,

$$\begin{aligned} s_m &= \sum_{n=1}^m (y_n - y_{n+1}) = \left(\sum_{n=1}^m y_n \right) - \left(\sum_{n=1}^m y_{n+1} \right) = [y_1 + y_2 + \cdots + y_m] - [y_2 + y_3 + \cdots + y_{m+1}] \\ &= y_1 - y_{m+1}. \end{aligned}$$

1.4 Geometric Series

Let k be a fixed integer and let $r \neq 0$ be a fixed real number. The infinite series $\sum_{n=k}^{\infty} r^n = r^k + r^{k+1} + r^{k+2} + \dots$ is called a **geometric series** with common ratio " r ". For example,

$$\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \text{ is a geometric series with common ratio } \frac{1}{2}.$$

Another example is that

$$\sum_{n=1}^{\infty} \frac{7^n}{29^n} \text{ is a geometric series with common ratio } \frac{7}{29}.$$

A non-example is the following:

$$\sum_{n=1}^{\infty} \frac{1}{n^2}.$$

We can easily find a formula for the m th partial sum of $\sum_{n=k}^{\infty} r^n$ where

$$\begin{aligned} s_1 &= r^k \\ s_2 &= r^k + r^{k+1} \\ s_3 &= r^k + r^{k+1} + r^{k+2} \\ &\vdots \\ s_m &= r^k + r^{k+1} + \dots + r^{k+m-1} \end{aligned} \quad (*)$$

Now, if $r = 1$, we have

$$s_m = \underbrace{1 + 1 + \dots + 1}_{m \text{ summands}} = m.$$

If $r \neq 1$, then multiply both sides of $(*)$ by r :

$$rs_m = r^{k+1} + r^{k+2} + \dots + r^{k+m}. \quad (**)$$

Subtracting $(**)$ from $(*)$, we get

$$s_m - rs_m = r^k - r^{k+m}.$$

Since $r \neq 1$, we have

$$s_m = \frac{r^k - r^{k+m}}{1 - r} = \frac{r^k(1 - r^m)}{1 - r}.$$

Note that

- (i) If $|r| < 1$, then $\lim_{m \rightarrow \infty} r^m = 0$.
- (ii) If $|r| > 1$ or $r = -1$, then $\lim_{m \rightarrow \infty} r^m$ does not exist.

Hence, we have

$$\lim_{m \rightarrow \infty} s_m = \begin{cases} \frac{r^k}{1-r} & \text{if } |r| < 1 \\ \text{DNE} & \text{if } |r| \geq 1. \end{cases}$$

Thus,

$$\sum_{n=k}^{\infty} r^n = \begin{cases} \frac{r^k}{1-r} & \text{if } |r| < 1 \\ \text{diverges} & \text{if } |r| \geq 1. \end{cases}$$

Example. • $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$

Observe that

$$\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n = \frac{\left(\frac{1}{2}\right)^1}{1 - \frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1.$$

• $\sum_{n=4}^{\infty} \left(\frac{1}{2}\right)^n$

Observe that

$$\sum_{n=4}^{\infty} \left(\frac{1}{2}\right)^n = \frac{\left(\frac{1}{2}\right)^4}{1 - \frac{1}{2}} = \frac{\left(\frac{1}{2}\right)^4}{\left(\frac{1}{2}\right)} = \frac{1}{8}.$$

1.5 Algebraic Limit Theorem for Series

Theorem. Let $(X, \|\cdot\|)$ be a normed space. Let (a_n) and (b_n) be two sequence in X . Suppose that

$$\sum_{n=1}^{\infty} a_n = A \quad (A \in X), \quad \sum_{n=1}^{\infty} b_n = B \quad (B \in X).$$

Then

- (i) For any scalar λ , $\sum_{n=1}^{\infty} (\lambda a_n) = \lambda A$.
- (ii) $\sum_{n=1}^{\infty} (a_n + b_n) = A + B$.

Proof. Can easily be proven via the Algebraic Limit Theorem for Sequences. ■

1.6 Divergence Test

Theorem (Divergence Test). Let $(X, \|\cdot\|)$ be a normed space. Let (x_n) be a sequence in X . If $\sum_{n=1}^{\infty} x_n$ converges, then $\lim_{n \rightarrow \infty} x_n = 0$.

Proof. Let $s_n = x_1 + \cdots + x_n$. Let $L = \sum_{n=1}^{\infty} x_n$. Note that

$$\sum_{n=1}^{\infty} x_n = L \implies \lim_{n \rightarrow \infty} s_n = L.$$

Also, note that

$$\forall n \geq 2 \quad x_n = s_n - s_{n-1}.$$

Note that $\lim s_n = L$ and $\lim s_{n-1} = L$. Therefore,

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} (s_n - s_{n-1}) = L - L = 0$$

by the Algebraic Limit Theorem for normed spaces. ■

Remark. Note that the divergence test is just the contrapositive of the above.

Example. • $\sum_{n=1}^{\infty} (-1)^n$ diverges because $\lim_{n \rightarrow \infty} (-1)^n$ does not exist.

• $\sum_{n=1}^{\infty} \frac{3n+1}{7n-4}$ diverges because $\lim_{n \rightarrow \infty} \frac{3n+1}{7n-4} = \frac{3}{7} \neq 0$.

From the above statements, we can now see make two key observations:

- If $\lim_{n \rightarrow \infty} x_n = 0$, then $\sum_{n=1}^{\infty} x_n$ may or may not converge.
- If $\lim_{n \rightarrow \infty} x_n \neq 0$, then $\sum_{n=1}^{\infty} x_n$ diverges.

As for the first observation above, we see that $\sum \frac{1}{n}$ diverges, but $\sum \frac{1}{n^2}$ converges.

1.7 Cauchy Criterion for Series

Theorem (Cauchy Criterion). Let $(X, \|\cdot\|)$ be a complete normed space. Let (x_n) be a sequence in X . Then

$$\sum_{k=1}^{\infty} x_k \text{ converges} \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall n > m > N \quad \left\| \sum_{k=1}^n x_k \right\| < \varepsilon.$$

Proof. Let $s_n = x_1 + \cdots + x_n$. Assuming that

$$\|s_n - s_m\| = \left\| \sum_{k=m+1}^n x_k \right\|$$

where $n > m$ and from the fact that

$$\begin{aligned} s_n - s_m &= (x_1 + \cdots + x_m + \cdots + x_n) - (x_1 + \cdots + x_m) \\ &= \sum_{k=m+1}^n x_k. \end{aligned}$$

Then we have

$$\begin{aligned} \sum_{k=1}^{\infty} x_k \text{ converges} &\iff (s_k) \text{ converges} \\ &\iff (s_k) \text{ is Cauchy} \\ &\iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall n > m > N \quad \|s_n - s_m\| < \varepsilon \\ &\iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall n > m > N \quad \left\| \sum_{k=m+1}^n x_k \right\| < \varepsilon \end{aligned}$$

as desired. ■

From here, we will refer to complete normed spaces as Banach spaces.

1.8 Absolute Convergence Test

Theorem (Absolute Convergence Test). Let $(X, \|\cdot\|)$ be a Banach Space. Let (x_n) be a sequence in X . If $\underbrace{\sum_{n=1}^{\infty} \|x_n\|}_{\text{a sum in } \mathbb{R}}$ converges, then $\sum_{n=1}^{\infty} x_n$ converges.

Proof. By the Cauchy Criterion for series, it suffices to show that

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall n > m > N \quad \left\| \sum_{k=m+1}^n x_k \right\| < \varepsilon. \quad (*)$$

Since $\sum_{k=1}^{\infty} \|x_k\|$ converges, and since $\mathbb{R}$ is complete, it follows from the Cauchy Criterion for series

that there exists $\hat{N}$ such that

$$\forall n > m > \hat{N} \quad \left| \sum_{k=m+1}^n \|x_k\| \right| < \varepsilon.$$

We claim that $\hat{N}$ is the same N we were looking for. Hence, if $n > m > \hat{N}$, then we have

$$\left\| \sum_{k=m+1}^n x_k \right\| \leq \sum_{k=m+1}^n \|x_k\| = \left| \sum_{k=m+1}^n \|x_k\| \right| < \varepsilon$$

as desired. ■

Please take note of the following observations:

- (1) If $\sum_{n=1}^{\infty} \|x_n\|$ converges, then $\sum_{n=1}^{\infty} x_n$ converges (in Banach spaces).
- (2) If $\sum_{n=1}^{\infty} \|x_n\|$ diverges, then $\sum_{n=1}^{\infty} x_n$ may converge or diverge.

From (2), we shall see (in the next lecture) that

- (1) $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{n} \right|$ diverges but $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges.
- (2) $\sum_{n=1}^{\infty} |(-1)^n|$ diverges, also $\sum_{n=1}^{\infty} (-1)^n$ diverges (by the divergence test).

Definition (Absolute Convergence and Conditional Convergence). We say that a series $\sum x_n$ **absolutely converges** if $\sum \|x_n\|$ converges and $\sum x_n$ converges. We say that $\sum x_n$ **conditionally converges** if $\sum \|x_n\|$ diverges but $\sum x_n$ converges.

Example (Conditionally Convergent). Consider $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$. We see that this series is conditionally convergent since

$$\left| \frac{(-1)^{n+1}}{n} \right| = \frac{1}{n} \rightarrow 0$$

which tell us that the above series diverges by the divergence test. But the above series converges via the Leibniz Test (As we shall see in the next lecture).

2 Lecture 21-22

2.1 Topics

- Cauchy Condensation Test
- Comparison Test
- More on lim sup and lim inf
- Root Test
- Ratio Test
- Dirichlet's Test

Theorem (Cauchy Condensation Test). Assume $a_n \geq 0$ for all n , and (a_n) is a decreasing sequence.

Then

$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \sum_{n=0}^{\infty} 2^n a_{2^n} = a_1 + 2a_2 + 4a_4 + 8a_8 + 16a_{16} + \cdots \text{ converges.}$$

Proof. Let $s_m = a_1 + \cdots + a_m$ and $t_m = a_1 + 2a_2 + 4a_4 + \cdots + 2^{m-1}a_{2^{m-1}}$. Using the fact that (a_n) is a decreasing sequence, we can see that

$$\begin{aligned} s_{2^k} &= a_1 = a_2 + (a_3 + a_4) + (a_5 + a_6 + a_7 + a_8) + \cdots + (a_{2^{k-1}+1} + \cdots + a_{2^k}) \\ &\geq a_1 + a_2 + (a_4 + a_4) + (a_8 + a_8 + a_8 + a_8) + \cdots + (a_{2^k} + \cdots + a_{2^k}) \\ &= a_1 + a_2 + 2a_4 + 4a_8 + \cdots + 2^{k-1}a_{2^k} \\ &= a_1 + \frac{1}{2}[t_{k+1} - a_1] \\ &= a_1 + \frac{1}{2}t_{k+1} - \frac{1}{2}a_1 \\ &= \frac{1}{2}(a_1 + t_{k+1}) \\ &\geq \frac{1}{2}t_{k+1}. \end{aligned}$$

Thus, we have

$$s_{2^k} \geq \frac{1}{2}t_{k+1}. \quad (*)$$

Similarly, we have

$$\begin{aligned} s_{2^{k-1}} &= a_1 + (a_2 + a_3) + (a_4 + a_5 + a_6 + a_7) + \cdots + (a_{2^{k-1}} + \cdots + a_{2^k-1}) \\ &\leq a_1 + (a_2 + a_2) + (a_4 + a_4 + a_4 + a_4) + \cdots + (a_{2^{k-1}} + \cdots + a_{2^k-1}) \\ &= a_1 + 2a_2 + 4a_4 + \cdots + 2^{k-1}a_{2^{k-1}} \\ &= t_k. \end{aligned}$$

Thus, we have that

$$s_{2^{k-1}} \leq t_k. \quad (**)$$

($\Leftarrow$) Assume that $\sum_{n=0}^{\infty} 2^n a_{2^n}$ converges ((t_m) converges). Our goal is to show that $\sum_{n=1}^{\infty} a_n$ converges; that is, (s_m) converges. Note that since $a_n \geq 0$, both (s_m) and (t_m) are increasing sequences. It follows from the Monotone Convergence Theorem that in order to prove (s_n) converges, it suffices to show that (s_m) is bounded.

Since (t_m) converges, we have that (t_m) is bounded. Hence, there exists $R > 0$ such that $t_m \leq R$ for all $m \in \mathbb{N}$. In what follows, we will show that R is an upper bound for (s_m) as well. Indeed, let $m \in \mathbb{N}$ be given. Choose k large enough so that $m < 2^k - 1$, then

$$s_m \leq s_{2^k-1} \leq t_k \leq R.$$

Thus, for all $m \in \mathbb{N}$, $0 \leq s_m \leq R$. Hence, (s_m) is bounded and so (s_m) converges by MCT.

($\Rightarrow$) Assume that $\sum_{n=1}^{\infty} a_n$ ((s_m) converges). Our goal is to show that $\sum_{n=0}^{\infty} 2^n a_{2^n}$ converges; that is, (t_m) converges.

We will prove the contrapositive: we will show that if (t_m) diverges then (s_m) diverges. Suppose (t_m) is divergent. Let $R > 0$ be given. We will show that there is a term in the nonnegative sequence (s_m) that is larger than R . Since (t_m) diverges and (t_m) is an increasing sequence (where $t_m \geq 0$), we see that (t_m) cannot be bounded above by the Monotone Convergence Theorem. Hence, there exists $k \in \mathbb{N}$ such that $t_{k+1} > 2R$. Now, we have

$$s_{2^k} \geq \frac{1}{2}t_{k+1} > \frac{1}{2}(2R) = R.$$

Thus, (s_m) is not bounded. ■

2.2 Applications of the Cauchy-Condensation Test

Example (P-series Test). Let $p > 0$. One can show that the sequence $\left(a_n = \frac{1}{n^p}\right)$ is a decreasing nonnegative sequence. Prove that

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges } \iff p > 1.$$

Proof. Using the Cauchy Condensation test, we have

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges } &\iff \sum_{n=0}^{\infty} 2^n \frac{1}{(2^n)^p} \text{ converges} \\ &\iff \sum_{n=0}^{\infty} \frac{1}{2^{np-n}} \text{ converges} \\ &\iff \sum_{n=0}^{\infty} \left(\frac{1}{2^{p-1}}\right)^n \text{ converges} \\ &\iff \left|\frac{1}{2^{p-1}}\right| < 1 \\ &\iff 1 < 2^{p-1} \\ &\iff 0 < p - 1 \\ &\iff 1 < p. \end{aligned}$$

In the fourth equivalence statement, we used the fact $\sum_{n=0}^{\infty} \left(\frac{1}{2^{p-1}}\right)^n$ is a geometric series with common ratio $\frac{1}{2^{p-1}}$. ■

Example. Let $p > 0$. One can show that the sequence $\left(a_n = \frac{1}{n(\ln n)^p}\right)_{n \geq 2}$ is a decreasing nonnegative sequence. Prove that

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p} \text{ converges } \iff p > 1.$$

Proof. By the Cauchy Condensation Test, we see that

$$\begin{aligned} \sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p} \text{ converges } &\iff \sum_{n=1}^{\infty} 2^n \frac{1}{2^n (\ln(2^n))^p} \text{ converges} \\ &\iff \sum_{n=1}^{\infty} \frac{1}{(n \ln 2)^p} \text{ converges} \\ &\iff \frac{1}{(\ln 2)^p} \sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges} \\ &\iff p > 1. \end{aligned}$$

Theorem (Comparison Test). Assume there exists an integer n_0 such that $0 \leq a_n \leq b_n$ for all $n \geq n_0$.

- (i) If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.
(ii) If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges.

Proof. Notice that (ii) is just the contrapositive of (i). So, it suffices to show (i).

By the Cauchy Criterion for convergence of series, it is enough to show that

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall n > m > N \quad \left| \sum_{k=m+1}^n a_k \right| < \varepsilon. \quad (*)$$

To this end, let $\varepsilon > 0$ be given. Our goal is to find an N such that $(*)$ holds. Since $\sum_{n=1}^{\infty} b_n$ converges, it follows from the Cauchy Criterion for series that

$$\exists \hat{N} \text{ such that } \forall n > m > \hat{N} \quad \left| \sum_{k=m+1}^n b_k \right| < \varepsilon.$$

Let $N = \max\{n_0, \hat{N}\}$. If $k \geq n_0$ where $a_k, b_k \geq 0$, we see that

$$\left| \sum_{k=m+1}^n a_k \right| = \sum_{k=m+1}^n a_k \text{ and } \left| \sum_{k=m+1}^n b_k \right| = \sum_{k=m+1}^n b_k. \quad (1)$$

Furthermore, if $k \geq n_0$, we have $a_k \leq b_k$, we have

$$\sum_{k=m+1}^n a_k \leq \sum_{k=m+1}^n b_k. \quad (2)$$

If $n > m > N$, we see that (1) and (2) imply that

$$\left| \sum_{k=m+1}^n a_k \right| = \sum_{k=m+1}^n a_k \leq \sum_{k=m+1}^n b_k = \left| \sum_{k=m+1}^n b_k \right| < \varepsilon.$$

■

Example. $(*)$ Does $\sum_{n=1}^{\infty} \frac{1}{n+5^n}$ converge?

Indeed, for all $n \in \mathbb{N}$, we have

$$0 \leq \frac{1}{n+5^n} \leq \frac{1}{5^n}.$$

Note that $\sum_{n=1}^{\infty} \frac{1}{5^n}$ converges (because it is a geometric series). Thus, the comparison test implies that $\sum_{n=1}^{\infty} \frac{1}{n+5^n}$ converges.

$(*)$ Suppose $a_n \geq 0$ and $\sum_{n=1}^{\infty} a_n$ converges. Prove that $\sum_{n=1}^{\infty} a_n^2$ converges.

Indeed, we see that

$$\sum_{n=1}^{\infty} a_n \text{ converges} \implies \lim_{n \rightarrow \infty} a_n = 0.$$

Hence, there exists $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $0 \leq a_n < 1$. Thus,

$$\forall n \geq n_0 \quad 0 \leq a_n^2 \leq a_n.$$

By the comparison test, we can conclude that $\sum_{n=1}^{\infty} a_n^2$ converges.

Remark (Some useful properties). Let (a_n) be a sequence of real numbers. Suppose $\lim_{n \rightarrow \infty} a_n = A \in \mathbb{R}$.

- (i) If $A < \beta$, then there exists $N \in \mathbb{N}$ such that for all $n > N$, $a_n < \beta$.
- (ii) If $\alpha < A$, then there exists N such that for all $n > N$, $\alpha < a_n$.

Theorem. Let (a_n) be a sequence of real numbers.

- (i) Suppose $\beta \in \mathbb{R}$ is such that $\limsup a_n < \beta$. Then

$$\exists N \in \mathbb{N} \text{ such that } \forall n > N \quad a_n < \beta.$$

- (ii) Suppose $\alpha \in \mathbb{R}$ is such that $\liminf a_n > \alpha$. Then

$$\exists N \in \mathbb{N} \text{ such that } \forall n > N \quad a_n > \alpha.$$

Proof. Here we will prove (i). Since $\limsup a_n < \beta$, we have $\limsup a_n \neq \infty$. We may consider two cases:

- (1) Suppose $\limsup a_n = -\infty$. Since $\liminf a_n \leq \limsup a_n$, we can conclude that $\liminf a_n = -\infty$. Therefore, $\lim a_n = -\infty$. The claim immediately follows from the definition of $a_n \rightarrow -\infty$.
- (2) Suppose $A = \limsup a_n$. Let $A = \limsup a_n$ and $r = \frac{\beta - A}{2}$. Since $\lim_{n \rightarrow \infty} \sup\{a_k : k \geq n\} = A$, there exists N such that

$$\forall n > N \quad \sup\{a_k : k \geq n\} < A + r.$$

In particular, we have

$$\forall n > N \quad \sup\{a_k : k \geq n\} < \beta.$$

Therefore, we have (noticing that $a_n \leq \sup\{a_k : k \geq n\}$),

$$\forall n > N \quad a_n < \beta.$$

Note that the proof of (ii) is completely analogous to the proof of (i). ■

Theorem. Let (a_n) be a sequence of real numbers.

- (i) Suppose $\limsup a_n > \beta$. Then, for infinitely many k , we have $a_k > \beta$. That is,

$$\forall n \in \mathbb{N} \quad \exists k \geq n \text{ such that } a_k > \beta.$$

- (ii) Suppose $\liminf a_n < \alpha$. Then, for infinitely many k , $a_k < \alpha$. That is,

$$\forall n \in \mathbb{N} \quad \exists k \geq n \text{ such that } a_k < \alpha.$$

Proof. Here we will prove (i) (the proof for (ii) is completely analogous). Assume for contradiction that only for finitely many k , $a_k > \beta$. Then there exists an $N \in \mathbb{N}$ such that for all $k > N$, we have $a_k \leq \beta$. Therefore,

$$\limsup a_k \leq \limsup \beta = \lim \beta = \beta$$

which contradicts the assumption that $\limsup a_k > \beta$. ■

Theorem (Root Test (Version 1)). Let (a_n) be a sequence of real numbers. Let $\alpha = \limsup \sqrt[n]{|a_n|}$.

- (i) If $\alpha < 1$, then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.
- (ii) If $\alpha > 1$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Theorem (Root Test (Version 2)). Let (a_n) be a sequence of real numbers. Assume that following limit exists:

$$\alpha = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$$

- (i) If $\alpha < 1$, then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.
- (ii) If $\alpha > 1$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Proof. (i) Choose a number β such that $\alpha < \beta < 1$. We have $\limsup \sqrt[n]{|a_n|} < \beta$. Then there exists an $N \in \mathbb{N}$ such that for all $n > N$, we have $\sqrt[n]{|a_n|} < \beta$. Hence, we have for all $n > N$, $0 \leq |a_n| < \beta^n$ and $\sum_{n=1}^{\infty} \beta^n$ converges (it is a geometric series with common ratio $0 < \beta < 1$). As a consequence, we see that $\sum_{n=1}^{\infty} |a_n|$ converges by the Comparison Test.

- (ii) Choose a number β such that $1 < \beta < \alpha$. We have $\beta < \limsup \sqrt[n]{|a_n|}$. By Useful Theorem 2, we have for all $n \in \mathbb{N}$, we have

$$\begin{aligned} \exists k \geq n \text{ such that } \sqrt[k]{|a_k|} > \beta &\implies |a_k| > \beta^k \\ &\implies \sup\{|a_m| : m \geq n\} > \beta^k. \end{aligned}$$

Since $k \geq n$, we have $\beta^k \geq \beta^n$, and so

$$\forall n \in \mathbb{N} \sup\{|a_m| : m \geq n\} > \beta^n.$$

Since $\lim_{n \rightarrow \infty} \beta^n = \infty$ ($\beta > 1$), it follows from the order limit theorem (for $\overline{\mathbb{R}}$) that $\lim_{n \rightarrow \infty} \sup\{|a_m| : m \geq n\} = \infty$. So, $\limsup |a_n| = \infty$. This tells us that $\lim a_n \neq 0$ (Explanation of Fact). So, $\sum a_n$ diverges by the Divergence Test. ■

Remark. This is just the contrapositive of the following fact:

$$\text{If } \lim a_n = 0, \text{ then } \lim |a_n| = 0, \text{ so } \limsup |a_n| = 0.$$

Theorem (Ratio Test (Version 1)). Let (a_n) be a sequence of real numbers.

- (i) If $\limsup \left| \frac{a_{n+1}}{a_n} \right| < 1$, then $\sum_{n=1}^{\infty} a_n$ converges absolutely.
- (ii) If $\left| \frac{a_{m+1}}{a_m} \right| \geq 1$ for all $n \geq n_0$ (some integer n_0), then $\sum_{n=1}^{\infty} a_n$ diverges.
- (iii) If $\liminf \left| \frac{a_{n+1}}{a_n} \right| \geq 1$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Theorem (Ratio Test (Version 2)). Let (a_n) be a sequence of real numbers. Assume that the following

limit exists:

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

- (i) If $\rho < 1$, then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent.
- (ii) If $\rho > 1$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Proof. (i) Choose a number β such that $\rho < \beta < 1$. We have

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \rho \implies \exists N \in \mathbb{N} \text{ such that } \forall n \geq N \quad \left| \frac{a_{n+1}}{a_n} \right| < \beta.$$

Thus, we have

$$\begin{aligned} |a_{N+1}| &< \beta |a_N| \\ |a_{N+2}| &< \beta |a_{N+1}| < \beta^2 |a_N| \\ |a_{N+3}| &< \beta |a_{N+2}| < \beta^3 |a_N| \\ &\vdots \end{aligned}$$

So, for all $n \in \mathbb{N}$, $|a_{N+n}| < \beta^n |a_N|$. Now, notice that

$$\sum_{n=1}^{\infty} \beta^n |a_N| = |a_N| \sum_{n=1}^{\infty} \beta^n \text{ converges.}$$

Now, notice that $\sum_{n=1}^{\infty} \beta^n |a_N| = |a_N| \sum_{n=1}^{\infty} \beta^n$ converges (since it is a geometric series with common ratio $0 < \beta < 1$). It follows from the Comparison Test that $\sum_{n=1}^{\infty} |a_{N+n}|$ converges. Considering that $\sum_{n=1}^{\infty} |a_{N+n}| = \sum_{n=N+1}^{\infty} |a_n|$, we can conclude that $\sum_{n=N+1}^{\infty} |a_n|$ converges. This immediately implies that $\sum_{n=1}^{\infty} |a_n|$ converges.

(ii) Choose a number β such that $1 < \beta < \rho$. Then we have

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \rho \implies \exists N \in \mathbb{N} \text{ such that } \forall n \geq N \quad \left| \frac{a_{n+1}}{a_n} \right| > \beta.$$

So, we have

$$\begin{aligned} |a_{N+1}| &> \beta |a_N| \\ |a_{N+2}| &> \beta |a_{N+1}| > \beta^2 |a_N| \\ |a_{N+3}| &> \beta |a_{N+2}| > \beta^3 |a_N| \\ &\vdots \end{aligned}$$

Thus, for each $n \in \mathbb{N}$, $|a_{N+n}| > \beta^n |a_N|$. Since $\beta > 1$, $\lim_{n \rightarrow \infty} \beta^n |a_N| = \infty$. So, $\lim_{n \rightarrow \infty} |a_{N+n}| = \infty$. Therefore, $\lim_{n \rightarrow \infty} a_{N+n} \neq 0$. Thus, $\lim_{n \rightarrow \infty} a_n \neq 0$ (because $(a_{N+n})_{n \geq 1}$ is a subsequence of $(a_n)_{n \geq 1}$). So, $\sum_{n=1}^{\infty} a_n$ diverges by the Divergence Test. ■

Example. Let $R \neq 0$ be a fixed number. Prove that the series $\sum_{n=1}^{\infty} \frac{R^n}{n!}$ converges. Indeed, we have

$$\begin{aligned} \rho &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{R^{n+1}}{(n+1)!}}{\frac{R^n}{n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{R^{n+1}n!}{R^n(n+1)!} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{R}{n+1} \right| \\ &= |R| \lim_{n \rightarrow \infty} \frac{1}{n+1} \\ &= 0. \end{aligned}$$

Thus, if $\rho = 0 < 1$, then $\sum_{n=1}^{\infty} \frac{R^n}{n!}$ is absolutely convergent. As a consequence, we have

$$\lim_{n \rightarrow \infty} \frac{R^n}{n!} = 0.$$

Remark. If (a_n) is a sequence and $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$, then $\lim_{n \rightarrow \infty} a_n = 0$.

Theorem (Dirichlet's Test). Let (b_n) be the sequence of partial sums of $\sum_{n=1}^{\infty} a_n$ be bounded, is a decreasing sequence of nonnegative numbers $(b_1 \geq b_2 \geq b_3 \geq \dots \geq 0)$, and $\lim_{n \rightarrow \infty} b_n = 0$. Then we have $\sum_{n=1}^{\infty} a_n b_n$ converges.

Corollary (Leibniz Test). Let (b_n) be a sequence in $\mathbb{R}$. Suppose $b_1 \geq b_2 \geq b_3 \geq \dots \geq 0$ and $\lim_{n \rightarrow \infty} b_n = 0$. Then we have $\sum_{n=1}^{\infty} (-1)^{n+1} b_n$ converges.

Consider the infinite sum

$$1 - 1 + \frac{1}{2} - \frac{1}{2} + \frac{1}{3} - \frac{1}{3} + \frac{1}{4} - \frac{1}{4} + \dots \quad (*)$$

and the following questions:

- (1) What is (s_n) ?
- (2) What is $\lim_{n \rightarrow \infty} s_n$?

Define the sequence of partial sums for the series in $(*)$ (s_n) .

Consider the following partial sums

$$s_1 = 1$$

$$s_2 = 1 - 1 = 0$$

$$s_3 = 1 - 1 + \frac{1}{2} = \frac{1}{2}$$

$$s_4 = 1 - 1 + \frac{1}{2} - \frac{1}{2} = 0$$

$$s_5 = 1 - 1 + \frac{1}{2} - \frac{1}{2} + \frac{1}{3} = \frac{1}{3}$$

$$\vdots$$

which establishes (1). Looking at the even subsequence of (s_n) , we can see that for all $k \in \mathbb{N}$ $s_{2k} = 0$ and the odd subsequence $s_{2k-1} = \frac{1}{k}$, respectively. Clearly, we can see from these subsequences that

$$s_{2k} \rightarrow 0 \text{ and } s_{2k-1} \rightarrow 0$$

as $k \rightarrow \infty$. Hence, we can see that $s_n \rightarrow 0$ which establishes (2).

2.3 Rearrangements

Consider the following rearrangement of (*)

$$1 + \frac{1}{2} - 1 + \frac{1}{3} + \frac{1}{4} - \frac{1}{2} + \frac{1}{5} + \frac{1}{6} - \frac{1}{3} + \cdots = \lim_{n \rightarrow \infty} s_n = \ln(2).$$

Consider the sequence of the partial sums of the above:

$$\begin{aligned} s_1 &= 1 \\ s_2 &= \frac{3}{2} \\ s_3 &= \frac{1}{2} \\ &\vdots \\ s_{2 \times 10^2 + 2} &\approx 0.6939 \\ s_{3 \times 10^4 + 2} &\approx 0.6932 \\ s_{3 \times 10^6 + 2} &\approx 0.6931 \\ &\vdots \end{aligned}$$

Theorem. If a series converges absolutely, then for any $L \in \mathbb{R}$, there exists some rearrangement of $\sum_{n=1}^{\infty} a_n$ converges to L .

$$\int_a^b f(x) \, dx$$